Centre for Development of Advanced Computing (C-DAC) Bangalore
Post Graduate Certificate Program in Advanced Computing (PGCP-AC)
August 2026 Batch
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Date: 2026-10-14
Assignment: 01
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1. Even / Odd Checker

Write a program to input a number and check whether it is even or odd.

2. Greatest of Three Numbers

Read three integers and print the largest.

3. Leap Year Checker

Input a year and determine whether it is a leap year.

4. Sum of First N Natural Numbers

Use a while loop to compute the sum of first N natural numbers.

5. Reverse a Number

Reverse a given integer (e.g., 123 → 321).

6. Prime Number Check

Check whether a number is prime.

7. Count Digits in a Number

Input a number and print how many digits it has.

8. Fibonacci Series (N terms)

Print the Fibonacci series up to N terms without recursion.

9. Factorial using Loop

Compute factorial of a number using a for loop.

10. Multiplication Table Generator

Input a number and print its table (10 rows).

11. Armstrong Number (for any number of digits)

Check whether a number is Armstrong sum of cube of digits is the number(e.g., 153, 9474).

12. Strong Number

Sum of factorial of digits equals the number.

13. Palindrome Number

Check if a number is the same when reversed.

14. Frequency of Each Digit

Print how many times each digit (0–9) occurs in a number.

15. Write a C++ program to calculate the electricity bill based on the number of units consumed. The bill should be calculated using the following slab rates:

Units Consumed	Rate per Unit
0–100 units	₹3.00
101–200 units	₹4.50
Above 200 units	₹6.00

An additional fixed charge of ₹50 is added to the electricity bill.

16. Write a C++ program to calculate the parking charges for a vehicle based on the

number of hours parked.

Parking Duration	Rate
First 2 hours	₹20 per hour
Next 3 hours	₹30 per hour
Beyond 5 hours	₹50 per hour

- Read the number of hours parked.
- Calculate parking charges progressively based on the duration.
- Add a fixed maintenance charge of ₹10.
- Display parking duration, parking charges, maintenance charge, and total amount payable.

17. Write a C++ program to implement a number guessing game.
- Store a secret number between 1 and 100.
- Allow the user a maximum of 7 attempts.
- After each incorrect guess, display whether the guessed number is higher or lower than the secret number.
- If the user guesses correctly, display the attempt number and terminate the loop using break.
- If all attempts are exhausted, display the secret number.

18. WAC++ program to print the following pattern based on number of lines entered by user:

```
*
* *
* * *
* * * *
```

19. WAC++ program to print the following pattern based on number of lines entered by user:

```
   *
  * *
 * * *
* * * *
```

20. WAC++ program to print the following pattern based on number of * in first line entered by user:

* * *
* *
*
* *
* * *